Properties of trapezoids

1

a $\overline{AD}$ and $\overline{BC}$

b $m\angle ABC = 60^\circ$

c $m\angle ADC = 131^\circ$

2

a $z = 26^\circ$

b $x = 85^\circ, y = 74^\circ$

c $x = 114, y = 114$

d $x = 90, y = 110$

3

a $m\angle B = 128^\circ$

b $m\angle C = 128^\circ$

c $m\angle D = 52^\circ$

4 $JL = 20$

5 $x = 13$

6 5

7

a $x = 69$

b $y = 2$

8

a $x = 12$

b 50°

9

a Yes

b No

c No

d No

10

a No

b No

c Yes

d No

11

To prove: Each pair of base angl[illegible] trapezoid $ABCD$ are congruent.

	Statements	Reasons
1.	$ABCD$ is an isosceles trapezoid	Given
2.	$\overline{AD} \cong \overline{BC}$	Definition of an isosceles trapezoid
3.	$\overline{AE} \cong \overline{BF}$	Altitudes of a trapezoid are congruent
4.	$\angle DEA$ and $\angle CFB$ are right angles	Definition of an altitude
5.	$\angle DEA \cong \angle CFB$	All right angles are congruent
6.	$\triangle AED \cong \triangle BFC$	Hypotenuse-leg congruence theorem
7.	$\angle ADE \cong \angle BCF$	Corresponding parts of congruent triangles are congruent (CPCTC)
8.	$\angle BCD$ and $\angle CBA$ are supplementary	Consecutive interior angles postulate
9.	$\angle ADC$ and $\angle BAD$ are supplementary	Consecutive interior angles postulate
10.	$\angle BAD \cong \angle ABC$	Congruent supplements theorem

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To prove: $\overline{AC} \cong \overline{BD}$

	Statements	Reasons
1.	$ABCD$ is an isosceles trapezoid	Given
2.	$\overline{AD} \cong \overline{BC}$	Definition of an isosceles trapezoid
3.	$\overline{DC} \cong \overline{DC}$	Reflexive property of congruence
4.	$\angle ADC \cong \angle BCD$	If a trapezoid is isosceles, then each pair of base angles is congruent.
5.	$\triangle ADC \cong \triangle BCD$	Side-angle-side congruence theorem
6.	$\overline{AC} \cong \overline{BD}$	Corresponding parts of congruent triangles are congruent (CPCTC)

Additional questions

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a $m\angle 1 = 77^\circ$
$m\angle 2 = 108^\circ$
$m\angle 3 = 108^\circ$

b $m\angle 1 = 69^\circ$
$m\angle 2 = 69^\circ$
$m\angle 3 = 111^\circ$

c $m\angle 1 = 120^\circ$
$m\angle 2 = 120^\circ$
$m\angle 3 = 60^\circ$

d $m\angle 1 = 105^\circ$
$m\angle 2 = 75^\circ$
$m\angle 3 = 75^\circ$

14 $FD = 3.5$

15 $FD = 1.26$

16 $x = 5^\circ$

17 $x = 7^\circ$

18 $x = 6$

19

a $x = 3$

b $BC = 27$

c $TU = 19.5$

d $KH = 12$

20 $x = 4$

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a True, since the legs of an isosceles trapezoid are congruent.

b True, since the diagonals of an isosceles trapezoid are congruent.

c False, the two bases of a trapezoid are not congruent.

d False, the legs of a trapezoid are not parallel.

e False, opposite angles in a trapezoid are not congruent.

f True, since the adjacent angles on the same leg of an isosceles trapezoid are congruent.

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To prov[redacted] $\cong \angle D$

	Statements	Reasons
1.	Trapezoid $ABCD$ with altitudes $\overline{BE}$ and $\overline{CF}$	Given
2.	$\overline{AB} \cong \overline{CD}$	Given
3.	$\overline{BE} \cong \overline{CF}$	Definition of altitude
4.	$\angle AEB, \angle CFD$ are right angles	Definition of altitude
5.	$\triangle AEB, \triangle CFD$ are right triangles	Definition of right triangle
6.	$\triangle AEB \cong \triangle CFD$ are right triangles	HL postulate
7.	$\angle A \cong \angle D$	CPCTC

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To prove: Trapezoid $ABCD$ is isosceles

	Statements	Reasons
1.	Trapezoid $ABCD$	Given
2.	$\overline{EF} \cong \overline{EG}$	Given
3.	$\overline{AF} \cong \overline{DG}$	Given
4.	$\overline{BG} \cong \overline{CF}$	Given
5.	$AF + FG = AG$	Segment addition postulate
6.	$DG + FG = FD$	Segment addition postulate
7.	$AF + FG = FD$	Substitution property
8.	$AG = FD$	Substitution property
9.	$\overline{AG} \cong \overline{FD}$	Definition of congruence
10.	$\triangle FEG$ is isosceles	Definition of isosceles triangle
11.	$\angle EFG \cong \angle EGF$	Definition of isosceles triangle
12.	$\triangle BAG \cong \triangle CDF$	SAS congruence theorem
13.	$\overline{BA} \cong \overline{CD}$	CPCTC
14.	Trapezoid $ABCD$ is isosceles	Definition of isosceles trapezoid

To prove: BKC[redacted] **parallelogram**

	Statements	Reasons
1.	Trapezoid $ABCD$	Given
2.	$\angle BAK \cong \angle BKA$	Given
3.	$\overline{KD} \| \overline{BC}$	Definition of isosceles trapezoid
4.	$\angle BKA \cong \angle KBC$	Converse of the alternate interior angles theorem
5.	$\angle CDK \cong \angle BAK$	Definition of isosceles trapezoid
6.	$\angle KBC \cong \angle CDK$	Transitive property
7.	$\angle KBC$ and $\angle BKD$ are supplementary angles	Converse of the consecutive interior angles theorem
8.	$\angle KBC + \angle BKD = 180^\circ$	Definition of supplementary angles
9.	$\angle BCD$ and $\angle CDK$ are supplementary angles	Converse of the consecutive interior angles theorem
10.	$\angle BCD + \angle CDK = 180^\circ$	Definition of supplementary angles
11.	$\angle KBC + \angle BKD = \angle BCD + \angle CDK$	Substitution property
12.	$\angle CDK + \angle BKD = \angle BCD + \angle CDK$	Substitution property
13.	$\angle BKD = \angle BCD$	Substraction property
14.	$\angle BKD \cong \angle BCD$	Definition of congruence
15.	$BKCD$ is a parallelogram	Definition of parallelogram